Module 5: Counting and Mathematical Induction

Counting principles, permutations, combinations,
pigeonhole principle, inclusion-exclusion and induction

Exam Focus

Scope. Syllabus 5.1 to 5.3: counting (basics of counting, pigeonhole principle), permutations and combinations, inclusion-exclusion principle, and mathematical induction.
Module 5 supplies the computational half of the paper. Almost every long question in the set ends with a Module 5 sub-part, so this module is where the marks are won or lost on arithmetic.

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Addition and multiplication principles

Addition principle

If a choice can be made in one of several mutually exclusive cases with $a_1, a_2, \ldots, a_k$ outcomes, the total number of outcomes is

$$a_1 + a_2 + \cdots + a_k.$$

If the cases overlap, remove the duplicate counts with inclusion-exclusion.

Multiplication principle

If a process has successive choices with $a_1, a_2, \ldots, a_k$ possibilities at each stage, the total number of outcomes is

$$a_1 a_2 \cdots a_k.$$

The number may vary by stage; multiply the actual choices available at each step.

Exam Focus

Worked examples. Three-letter strings followed by three digits, repetition allowed: $26^3 \cdot 10^3$. Select one boy from 28 and one girl from 18: $28 \cdot 18 = 504$. A die rolled five times: $6^5 = 7776$.

Past-paper question

Nov 2024 Section A Q6 and Oct 2025 Q5(a)(ii). Explain the multiplication rule of counting.

Solution

If an event can occur in m_1 ways at the first stage, m_2 ways at the second stage, and so on through m_r stages, and the choice at each stage is independent of the earlier choices, then the total number of ways of performing the whole sequence of operations is

$$m_1 \times m_2 \times \cdots \times m_r.$$

For example, if a boy or a girl is chosen and then a prize is selected, the total number of ways is $2 \times n$, not $2 + n$, because every person can receive every prize.

Permutations

An arrangement of n distinct objects taken r at a time is a permutation:

$${}^nP_r = P(n, r) = \frac{n!}{(n-r)!}.$$

If repetition is allowed and each of r positions has n choices, there are n^r strings.

If all n objects are arranged with repeated indistinguishable types of multiplicities $n_1, \ldots, n_k$, the number of distinct arrangements is

$$\frac{n!}{n_1! n_2! \cdots n_k!}.$$

Exam Focus

Worked examples. Four distinct letters selected and ordered from six: $6P4 = 6!/2! = 360$. Arrangements of ROOTS: $5!/2! = 60$ (two O's). A six-digit PIN from digits 0–9 with repetition: 10^6; a leading zero is allowed for a PIN.

Past-paper question

Oct 2024 CAE-2 Q1B(b). In how many ways can the letters of the word ATTITUDE be rearranged such that no two T's are adjacent to each other?

Solution

ATTITUDE has 8 letters: A, T, T, I, T, U, D, E. Three T's and five other distinct letters.
Method (gap method). First arrange the five non-T letters A, I, U, D, E in a row: $5!$ ways. This creates 6 gaps:

$$_\,L_1\,_\,L_2\,_\,L_3\,_\,L_4\,_\,L_5\,_$$

To keep the T's apart, put at most one T in each gap. Choose 3 of the 6 gaps: $\binom{6}{3} = 20$.

$$\text{Answer} = 5!\binom{6}{3} = 120 \times 20 = \boxed{2400}.$$

Check. Total distinct arrangements without restriction: $\frac{8!}{3!} = 6720$. So 2400 of 6720 have no two T's adjacent, which is a sensible fraction.

Past-paper question

Oct 2025 Q5(a)(iii). Find the number of permutations of the letters of the word ALLAHABAD.

Solution

ALLAHABAD has 9 letters: A appears 4 times, L appears 2 times, and H, B, D each appear once.

$$\text{Answer} = \frac{9!}{4!\,2!} = \frac{362880}{48} = \boxed{7560}.$$

Past-paper question

Oct 2025 Q5(b)(iii). In how many different ways can the letters of the word LEADING be arranged in such a way that the vowels always come together?

Solution

LEADING has 7 distinct letters, of which E, A and I are vowels.
Treat the three vowels as a single block. Then we have 5 objects to arrange: the block $[EAI]$, and L, D, N, G. These can be arranged in $5!$ ways. Within the block the three vowels can be ordered in $3!$ ways.

$$\text{Answer} = 5! \times 3! = 120 \times 6 = \boxed{720}.$$

Do Not Confuse

In the block method you must multiply by the internal arrangements of the block. $5!$ alone would count only one particular vowel order inside the block.

Combinations

A selection where order does not matter is a combination:

$${}^nC_r = C(n,r) = \frac{n!}{r!(n-r)!}, \qquad {}^nP_r = {}^nC_r\, r!.$$

Useful identities:

$$\binom{n}{0} = \binom{n}{n} = 1, \qquad \binom{n}{1} = n, \qquad \binom{n}{r} = \binom{n}{n-r},$$

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}, \qquad \sum_{r=0}^{n} \binom{n}{r} = 2^n.$$

Exam Focus

Worked examples. Teams of 3 from 8: $\binom{8}{3} = 56$. A committee of 3 from 6 faculty including both named professors: choose the third member from the other four, so 4. From 5 boys and 4 girls, 3-person committees with at least one girl: $\binom{9}{3} - \binom{5}{3} = 84 - 10 = 74$.

Do Not Confuse

Use permutations when order or role assignment matters; use combinations when only membership matters. A PIN permits a leading zero; a number usually does not.

Past-paper question

Nov 2024 Q17(a)(i) and Oct 2025 Q5(a)(i). From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

Solution

Split into cases by the number of men:

$$\text{3 men, 2 women} : \binom{7}{3}\binom{6}{2} = 35 \times 15 = 525,$$
$$\text{4 men, 1 woman} : \binom{7}{4}\binom{6}{1} = 35 \times 6 = 210,$$
$$\text{5 men, 0 women} : \binom{7}{5}\binom{6}{0} = 21 \times 1 = 21.$$

$$\text{Total} = 525 + 210 + 21 = \boxed{756}.$$

Check by complement. Total committees of 5 from 13 people: $\binom{13}{5} = 1287$. Committees with fewer than 3 men: 0 men $\binom{6}{0}\binom{6}{5} = 6$, 1 man $\binom{7}{1}\binom{6}{4} = 105$, 2 men $\binom{7}{2}\binom{6}{3} = 420$. So $1287 - (6 + 105 + 420) = 1287 - 531 = 756$. Same answer.

Past-paper question

Nov 2024 Q17(a)(ii) and Oct 2025 Q5(a)(ii). How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

Solution

A number is divisible by 5 when its last digit is 0 or 5. The only choice of 5 from the given digits is 5, so the last digit must be 5.
The first two digits come from $\{2, 3, 6, 7, 9\}$, chosen without repetition and in order:

$$5 \times 4 = 20.$$

$$\text{Answer} = \boxed{20}.$$

Past-paper question

Oct 2025 Q5(b)(ii). How many 6-digit numbers can be formed by using the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 if every number is to start with '30' with no digit repeated?

Solution

The first two digits are fixed as 3 and 0. The remaining four places are filled from $\{1, 2, 4, 5, 6, 7, 8\}$, seven digits, without repetition, in order:

$$7 \times 6 \times 5 \times 4 = \boxed{840}.$$

Exam Focus

Worked, addition rule for sweets. A bakery has 15 different cupcakes, 20 different doughnuts and 13 different muffins. How many different choices of sweets can you choose from? The choices are mutually exclusive, so add:

$$15 + 20 + 13 = \boxed{48}.$$

Pigeonhole principle

Basic principle: placing N objects into k boxes with $N > k$ guarantees a box with at least two objects.

Generalized principle: some box contains at least $\lceil \frac{N}{k} \rceil$ objects.

To guarantee at least r objects in one of k boxes, the minimum required is

$$k(r-1) + 1.$$

Past-paper question

Oct 2024 CAE-2 Q1A(b). Explain the pigeonhole principle with a suitable example.

Solution

Statement. If N objects are placed into k boxes and $N > k$, then at least one box contains at least two objects. More generally, if $N > k(r-1)$, then at least one box contains at least r objects; equivalently some box contains at least $\lceil N/k \rceil$ objects.
Example. Among 47 students, 8 groups must be formed. Since $\lceil 47/8 \rceil = 7$, some group contains at least 7 students.
Another example. To guarantee that at least 4 people in a group share the same day of the week, we need $7(4-1)+1 = 22$ people.

Inclusion-exclusion principle

For two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

For *exactly two* of the three sets, add the three pairwise intersections and subtract three times the triple intersection. For *exactly one*, add the single-set totals, subtract twice the pairwise intersections, then add three times the triple intersection.

Exam Focus

Worked examples. Of 80 students, 50 use Android and 45 iPhone, and everyone uses at least one: both $= 50 + 45 - 80 = 15$. For the language survey in the Oct 2024 paper, at least one $= 50 + 45 + 40 - 20 - 18 - 15 + 8 = 90$.

Past-paper question

Oct 2024 CAE-2 Q1A(a). In a class of students undergoing a computer course: of a total of 50 students, 30 know Pascal, 18 know Fortran, 26 know COBOL, 9 know both Pascal and Fortran, 16 know both Pascal and COBOL, 8 know both Fortran and COBOL, and 47 know at least one of the three. Determine (i) how many students know none of these languages, and (ii) how many know all three.

Solution

Let P, F, C be the three sets.
(i) None $= 50 - 47 = \boxed{3}$.
(ii) Let $t = |P \cap F \cap C|$. By inclusion-exclusion,

$$47 = 30 + 18 + 26 - 9 - 16 - 8 + t = 41 + t,$$

so $t = \boxed{6}$.
Useful derived counts: exactly one language $= 74 - 2(33) + 3(6) = 26$; exactly two $= (9-6) + (16-6) + (8-6) = 15$.

Past-paper question

Oct 2024 CAE-2 Q1B(a). Among 150 college students, 83 own cars, 97 own bike, 28 own motorcycles, 53 own a car and a bike, 14 own a car and a motorcycle, 7 own a bike and a motorcycle, and 2 own all three. (a) How many students do not own any of the three? (b) How many students own a bike and nothing else?

Solution

(a) By inclusion-exclusion the union is

$$83 + 97 + 28 - 53 - 14 - 7 + 2 = 136.$$

So none $= 150 - 136 = \boxed{14}$.
(b) Bike only $= 97 - 53 - 7 + 2 = \boxed{39}$.
Exactly one of the three: use the "exactly one" variant of the formula,

$$\text{exactly one} = (83 + 97 + 28) - 2(53 + 14 + 7) + 3(2) = 208 - 148 + 6 = 66.$$

Exactly two: $(53 + 14 + 7) - 3(2) = 74 - 6 = 68$. Check: $66 + 68 + 2 + 14 = 150$, which is the total number of students.

Past-paper question

Nov 2024 Q17(b). A software company employs 100 programmers. 45 are proficient in Java, 30 in C#, 20 in Python, 6 in C# and Java, 1 in Java and Python, 5 in C# and Python, and just 1 is proficient in all three. Determine the number of programmers not proficient in any of these three languages.

Solution

Union by inclusion-exclusion:

$$45 + 30 + 20 - 6 - 1 - 5 + 1 = 84.$$

Not proficient in any:

$$100 - 84 = \boxed{16}.$$

Mathematical induction

To prove $P(n)$ for all integers $n \geq n_0$:

1. State $P(n)$ and the starting value n_0.

2. **Base case:** verify $P(n_0)$ directly.
3. **Inductive hypothesis:** assume $P(k)$ is true for an arbitrary $k \geq n_0$.
4. **Inductive step:** use that assumption to prove $P(k+1)$.
5. Conclude by induction that $P(n)$ holds for every integer $n \geq n_0$.

Do Not Confuse

In the inductive step, assume the claim only for k, then derive the $k+1$ case. Do not assume the statement you are trying to prove for $k+1$. Include both the base case and the final conclusion.

Exam Focus

Worked proof. Prove $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for $n \geq 1$.
Base case: $n = 1$ gives $1 = \frac{1 \cdot 2}{2}$. True.
Inductive hypothesis: assume $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$ for some $k \geq 1$.
Inductive step:

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1)\left(\frac{k}{2} + 1\right) = \frac{(k+1)(k+2)}{2}.$$

This is exactly the formula for $n = k+1$.
Conclusion: by mathematical induction, $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 1$.

Past-paper question

Oct 2025 Q5(b)(i). Using mathematical induction, prove that $3^n - 1$ is a multiple of 2 for $n = 1, 2, \ldots$

Solution

Base case: $n = 1$. $3^1 - 1 = 3 - 1 = 2$, which is a multiple of 2. True.
Inductive hypothesis: assume $3^k - 1$ is a multiple of 2, that is $3^k - 1 = 2m$ for some integer m.
Inductive step: consider $3^{k+1} - 1$:

$$3^{k+1} - 1 = 3 \cdot 3^k - 1 = 3(3^k - 1) + 2 = 3(2m) + 2 = 2(3m + 1).$$

Since $3m + 1$ is an integer, $3^{k+1} - 1$ is a multiple of 2.
Conclusion: by mathematical induction, $3^n - 1$ is a multiple of 2 for all $n \geq 1$.

Do Not Confuse

The printed question in the Oct 2025 paper reads "$3n - 1$ is a multiple of 2". As printed that is false (for $n = 2$ it gives 5). The intended statement is $3^n - 1$, which is what the proof above establishes and what the standard textbook version of this question asks. Write the exponent clearly and mention the reading in your answer if you are unsure.

Past-paper question

Oct 2024 CAE-2 Q2B(a). Using mathematical induction, prove the given statement: for any natural number n, $2^{2n} - 1$ is divisible by 3.

Solution

Base case: $n = 1$. $2^2 - 1 = 4 - 1 = 3$, which is divisible by 3. True.
Inductive hypothesis: assume $2^{2k} - 1$ is divisible by 3, so $2^{2k} - 1 = 3m$ for some integer m.
Inductive step:

$$2^{2(k+1)} - 1 = 2^{2k+2} - 1 = 4 \cdot 2^{2k} - 1 = 4(2^{2k} - 1) + 3 = 4(3m) + 3 = 3(4m + 1).$$

Since $4m + 1$ is an integer, $2^{2(k+1)} - 1$ is divisible by 3.
Conclusion: by mathematical induction, $2^{2n} - 1$ is divisible by 3 for all $n \geq 1$.

Exam Focus

Worked proof, sum of odd numbers. Prove $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for $n \geq 1$.
Base case: $n = 1$ gives $1 = 1^2$. True.
Inductive hypothesis: assume $1 + 3 + \cdots + (2k - 1) = k^2$.
Inductive step:

$$1 + 3 + \cdots + (2k - 1) + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2,$$

which is the formula for $n = k + 1$.
Conclusion: by mathematical induction, $1 + 3 + \cdots + (2n - 1) = n^2$ for all $n \geq 1$.

Quick method selector

Clue in question	Method
"either ... or" in disjoint cases	Addition rule
several stages / "and then"	Multiplication rule
arrange, order, rank, positions	Permutation
choose, select, committee, team	Combination
guarantee a repeated category	Pigeonhole principle
overlapping groups / at least one set	Inclusion-exclusion
prove for every natural number	Mathematical induction
arrange letters of a word	Permutation with repeated letters
no two of a letter together	Gap method

Answer skeletons

Answer Skeleton

Arrangement of letters (3 to 9 marks). Count the letters and the multiplicities. Decide whether order within a group matters (it does). Write $n!$ divided by the factorials of the repeated counts. State the answer.

Answer Skeleton

Selection with a restriction (9 marks). Split into mutually exclusive cases by the number from each group. In each case write the combination product. Add the cases. Verify with a complement count if time allows.

Answer Skeleton

Digit formation (3 to 9 marks). Identify the fixed positions first, then the last-digit divisibility rule, then count the free positions as an ordered selection without repetition.

Answer Skeleton

Inclusion-exclusion (9 marks). Label the sets, write the formula with all seven terms, substitute, and simplify. Then answer the specific sub-question (none, exactly one, exactly two) with the corresponding variant of the formula.

Answer Skeleton

Prove by induction (5 to 9 marks). Let $P(n)$ be the claim. Verify the base case. Assume $P(k)$. Starting from the left side for $k + 1$, use the inductive hypothesis and simplify to the required right side. State that the claim follows for all specified n.

Self-test

1. State the addition and multiplication principles of counting.
2. How many 4-digit numbers can be made from the digits 1–9 without repetition?

3. How many arrangements are there of the letters of ROOTS? Of INDEPENDENCE? How many of INDEPENDENCE start with P?
4. In how many ways can the letters of ATTITUDE be rearranged so that no two T's are adjacent?
5. How many arrangements are there of the letters of ALLAHABAD?
6. In how many ways can the letters of LEADING be arranged so that the vowels are together?
7. From 6 men and 5 women, how many 4-person committees include at least one woman?
8. From a group of 7 men and 6 women, how many 5-person committees have at least 3 men?
9. How many 3-digit numbers from 2, 3, 5, 6, 7, 9 are divisible by 5 with no digit repeated?
10. How many integers from 1 to 100 are divisible by 2 or 5?
11. What is the minimum number of people that guarantees at least 4 were born on the same day of the week?
12. State the generalized pigeonhole principle and the two-set inclusion-exclusion formula.
13. Prove by induction that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for $n \geq 1$.
14. Prove by induction that $3 \mid (n^3 + 2n)$ for every positive integer n.

Answers

1. Addition: for mutually exclusive cases, add the counts. Multiplication: for successive independent stages, multiply the counts.
2. $9 \times 8 \times 7 \times 6 = \boxed{3024}$.
3. ROOTS: $\frac{5!}{2!} = \boxed{60}$. INDEPENDENCE (12 letters, E four times, N three times, D twice): $\frac{12!}{4!\,3!\,2!} = \boxed{1663200}$. Starting with P: $\frac{11!}{4!\,3!\,2!} = \boxed{138600}$.
4. $5!\binom{6}{3} = \boxed{2400}$.
5. $\frac{9!}{4!\,2!} = \boxed{7560}$.
6. $5! \times 3! = \boxed{720}$.
7. $\binom{11}{4} - \binom{6}{4} = 330 - 15 = \boxed{315}$.
8. $\binom{7}{3}\binom{6}{2} + \binom{7}{4}\binom{6}{1} + \binom{7}{5}\binom{6}{0} = 525 + 210 + 21 = \boxed{756}$.
9. $5 \times 4 = \boxed{20}$.
10. $50 + 20 - 10 = \boxed{60}$.
11. $7(4 - 1) + 1 = \boxed{22}$.
12. Some box contains at least $\lceil N/k \rceil$ objects. For two sets, $|A \cup B| = |A| + |B| - |A \cap B|$.
13. Base case $n = 1$: $1 = 1^2$. Assume $1 + 3 + \cdots + (2k - 1) = k^2$. Then $1 + 3 + \cdots + (2k - 1) + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$. Hence the result holds for all $n \geq 1$.
14. Base case $n = 1$: $1 + 2 = 3$, divisible by 3. Assume $k^3 + 2k = 3m$. Then $(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 3k + 1 + 2k + 2 = (k^3 + 2k) + 3k^2 + 5k + 3 = 3m + 3(k^2 + k + 1) = 3(m + k^2 + k + 1)$. Hence $3 \mid (n^3 + 2n)$ for all positive integers n.

Last-minute page

PRINCIPLES
Addition: sum disjoint cases
Multiplication: product of stages

PERMUTATIONS
${}^nP_r = \dfrac{n!}{(n-r)!}$
Repetition allowed: n^r
Repeated letters: $\dfrac{n!}{\prod n_i!}$

COMBINATIONS
${}^nC_r = \dfrac{n!}{r!(n-r)!}$
${}^nP_r = {}^nC_r\, r!$
$\sum_{r=0}^{n} \binom{n}{r} = 2^n$

GUARANTEES / OVERLAP
Pigeonhole: at least $\lceil N/k \rceil$
Guarantee r: $k(r-1)+1$
$|A \cup B| = |A| + |B| - |A \cap B|$
Three sets: singles $-$ pairs $+$ triple

SPECIAL CASES
No two T's adjacent: $5!\binom{6}{3}$
Vowels together: $5! \times 3!$
Gap method for repeated letters

INDUCTION
Base case $\rightarrow$ assume $P(k) \rightarrow$ prove $P(k+1) \rightarrow$ conclude

DO NOT CONFUSE
Order matters: permutation; no order: combination.
A PIN allows a leading zero; a number does not.
In a block method, multiply by the block's internal arrangements.

Identify the rule, show the working, and state the conclusion. Split restricted selections into explicit cases.